

Small Business Insurability Evaluation System: Feature Specification

Overview

This document outlines the data model for a small business insurability evaluation system. It provides a condensed list of **Raw Features** (data points collected directly from the user or uploaded files) and **Derived Features** (calculated or inferred metrics based on the raw data). This structured approach enables the system to assess a business's readiness for insurance and estimate potential risk profiles.

1. Raw Features

These are the fundamental data points the system needs to ingest. They can be gathered via user questionnaires, form uploads (like the ACORD 125), or API integrations.

1.1 Business Identity & Operations

Feature Name	Description	Data Type	Source Example
business_name	Legal name of the entity.	String	User Input / ACORD 125
entity_type	Legal structure (e.g., LLC, Corporation, Sole Proprietorship).	Categorical	User Input / ACORD 125
years_in_business	Number of years the business has been operating.	Integer	User Input / ACORD 125
naics_code	6-digit North American Industry Classification System code.	String	User Input / Tax Docs
sic_code	4-digit Standard Industrial Classification code (legacy).	String	User Input / Tax Docs

description_of_operations	Free-text description of primary business activities.	Text	User Input / ACORD 125
physical_address	Primary physical address of the business.	String	User Input / ACORD 125
mailing_address	Mailing address of the business (if different from physical).	String	User Input / ACORD 125
contact_phone	Primary contact phone number.	String	User Input / ACORD 125
contact_email	Primary contact email address.	String	User Input / ACORD 125
business_website	Official business website URL.	String	User Input / ACORD 125
fein	Federal Employer Identification Number.	String	User Input / ACORD 125
number_of_members	Number of members/partners/officers in the business.	Integer	User Input / ACORD 125
additional_named_insureds	List of additional entities/individuals to be named as insureds.	List of Strings	User Input / ACORD 125

1.2 Financial & Employee Data

Feature Name	Description	Data Type	Source Example
annual_revenue	Total gross revenue for the most recent fiscal year.	Float	Financial Statements
total_payroll	Total annual payroll expenses.	Float	Payroll Records

employee_count_ft	Number of full-time employees.	Integer	Payroll Records
employee_count_pt	Number of part-time or seasonal employees.	Integer	Payroll Records
credit_score	Business credit score (if available).	Integer	Credit Report
sales_percentage_installation_service	Percentage of sales derived from installation or service activities.	Float	User Input / Operations Details
payroll_by_classification	Detailed breakdown of payroll by job classification.	JSON/Object	Payroll Records / ACORD 125

1.3 Claims & Loss History (Last 3-5 Years)

Feature Name	Description	Data Type	Source Example
total_claims_count	Total number of insurance claims filed.	Integer	Loss Runs
total_claims_paid	Total dollar amount paid out by insurers for past claims.	Float	Loss Runs
open_claims_count	Number of claims currently open or litigated.	Integer	Loss Runs
prior_coverage_declined	Has the business ever been declined coverage? (Yes/No)	Boolean	User Input
claims_dates_of_loss	Dates of each loss event.	List of Dates	Loss Runs
claims_description	Brief description of each claim event.	List of Strings	Loss Runs

claims_reserved_amounts	Funds set aside for potential future payments for each claim.	List of Floats	Loss Runs
claims_status	Status of each claim (e.g., Open, Closed, Litigated).	List of Categorical	Loss Runs
prior_carrier_name	Name of previous insurance carrier.	String	Prior Carrier Info
prior_policy_number	Policy number with previous carrier.	String	Prior Carrier Info
prior_policy_category	Type of policy with previous carrier.	Categorical	Prior Carrier Info
prior_policy_dates	Effective and expiration dates of prior policy.	Date Range	Prior Carrier Info
prior_policy_premium	Premium paid for prior policy.	Float	Prior Carrier Info
prior_underwriting_questions	Answers to specific underwriting questions from prior applications.	JSON/Object	Prior Carrier Info

1.4 Property & Location Data

Feature Name	Description	Data Type	Source Example
primary_zip_code	ZIP code of the primary business location.	String	User Input
building_construction_type	Material type (e.g., Frame, Masonry, Fire-Resistive).	Categorical	Property Inspection
building_year_built	Year the primary structure was built.	Integer	Property Records
building_year_updated	Year of last major structural/system	Integer	User Input

	update.		
fire_alarm_present	Is a central station fire alarm installed? (Yes/No)	Boolean	User Input
sprinkler_system_present	Is a functional sprinkler system installed? (Yes/No)	Boolean	User Input
square_footage	Total square footage of the primary business premises.	Float	Property Inspection / User Input
leased_area	Is the business premises leased? (Yes/No)	Boolean	User Input
fire_extinguishers_present	Are fire extinguishers present and regularly maintained? (Yes/No)	Boolean	User Input
security_system_present	Is a security system installed? (Yes/No)	Boolean	User Input
distance_to_fire_station	Distance to the nearest fire station in miles.	Float	User Input / GIS Data
distance_to_fire_hydrant	Distance to the nearest fire hydrant in feet.	Float	User Input / GIS Data

1.5 Cybersecurity & Safety

Feature Name	Description	Data Type	Source Example
formal_safety_program	Does the business have a documented safety program? (Yes/No)	Boolean	User Input

mfa_implemented	Is Multi-Factor Authentication used for system access? (Yes/No)	Boolean	IT Questionnaire
data_backups_regular	Are regular, off-site data backups performed? (Yes/No)	Boolean	IT Questionnaire
employee_safety_training	Is regular employee safety training conducted? (Yes/No)	Boolean	User Input
osha_compliance	Is the business compliant with OSHA regulations? (Yes/No)	Boolean	User Input
incident_response_plan	Does the business have a documented incident response plan? (Yes/No)	Boolean	IT Questionnaire
third_party_vendor_risk_management	Does the business assess cybersecurity risks of third-party vendors? (Yes/No)	Boolean	IT Questionnaire

2. Derived Features

These features are calculated by the system using the raw data. They provide higher-level insights into the business's risk profile and insurability readiness.

2.1 Operational Risk Metrics

Derived Feature	Description	Derived From (Raw Features)	Calculation / Logic
industry_risk_tier	Categorizes the business into Low, Medium, or High risk based on industry.	naics_code , sic_code	Map NAICS/SIC to a predefined risk matrix (e.g., Construction = High, Consulting = Low).

business_maturity_score	Assesses stability based on longevity.	years_in_business	e.g., < 1 year = High Risk; 1-3 years = Medium Risk; > 3 years = Low Risk.
revenue_per_employee	Indicates operational efficiency and potential exposure scale.	annual_revenue , employee_count_ft , employee_count_pt	$\text{annual_revenue} / (\text{employee_count_ft} + (0.5 * \text{employee_count_pt}))$
total_employee_count	Total number of employees (full-time + part-time).	employee_count_ft , employee_count_pt	$\text{employee_count_ft} + \text{employee_count_pt}$
operational_complexity_score	Score indicating the complexity of business operations.	description_of_operations , sales_percentage_installation_service , number_of_members , additional_named_insureds	Qualitative assessment based on text analysis of operations, presence of installation/service, multiple members/insureds.

2.2 Claims & Financial Risk Metrics

Derived Feature	Description	Derived From (Raw Features)	Calculation / Logic
claim_frequency_rate	Average number of claims per year.	total_claims_count , years_in_business (or specific loss run period)	$\text{total_claims_count} / [\text{Years of Loss Run Data}]$
average_claim_severity	Average cost per claim.	total_claims_paid , total_claims_count	$\text{total_claims_paid} / \text{total_claims_count}$ (if count > 0).
loss_ratio_estimate	Estimated ratio of losses to premium (requires estimated premium).	total_claims_paid , [Estimated Premium]	$\text{total_claims_paid} / [\text{Estimated Premium}]$. High ratio indicates poor insurability.

financial_stability_flag	Flags potential financial distress.	credit_score , prior_coverage_declined	True if credit_score < threshold OR prior_coverage_declined is True.
claims_open_ratio	Ratio of open claims to total claims.	open_claims_count , total_claims_count	open_claims_count / total_claims_count (if total_claims_count > 0).
prior_insurance_stability	Indicates consistency of prior insurance coverage.	prior_carrier_name , prior_policy_dates	Assess gaps in coverage or frequent carrier changes.

2.3 Property & Location Risk Metrics

Derived Feature	Description	Derived From (Raw Features)	Calculation / Logic
effective_building_age	Adjusted age considering updates.	building_year_built , building_year_updated	Current Year - building_year_updated (if updated), else Current Year - building_year_built .
property_protection_score	Aggregate score of physical safety measures.	building_construction_type , fire_alarm_present , sprinkler_system_present	Weighted sum: Fire-Resistive (+3), Sprinklers (+2), Alarms (+1). Higher is better.
location_hazard_index	External risk score based on geography.	primary_zip_code	API call to external service (e.g., FEMA, crime databases) using ZIP code to return a 1-10 risk score.
fire_protection_rating	Combined score for fire safety measures.	fire_alarm_present , sprinkler_system_present , fire_extinguishers_present , distance_to_fire_station ,	Weighted score based on presence of systems and proximity to emergency services.

		distance_to_fire_hydrant	
premises_ownership_status	Indicates if the business owns or leases its primary premises.	leased_area	"Owned" if leased_area is False, "Leased" if True.

2.4 Cyber & Liability Readiness

Derived Feature	Description	Derived From (Raw Features)	Calculation / Logic
cyber_readiness_score	Assessment of basic IT security.	mfa_implemented , data_backups_regular	Boolean AND logic or weighted score. If both True, readiness is High.
safety_culture_indicator	Assessment of physical safety commitment.	formal_safety_program , claim_frequency_rate , employee_safety_training , osha_compliance	High if formal_safety_program , employee_safety_training , osha_compliance are True AND claim_frequency_rate is Low.
cyber_risk_posture	Comprehensive assessment of cybersecurity measures.	mfa_implemented , data_backups_regular , incident_response_plan , third_party_vendor_risk_management	Weighted score based on implementation of various cybersecurity controls.

Conclusion

By structuring the ingestion system to capture these specific **Raw Features**, the platform can automatically generate the **Derived Features**. This derived data provides a comprehensive, standardized profile of a small business, allowing for rapid evaluation of their insurability, identification of risk mitigation opportunities, and more accurate premium estimations.